



Investigating the Effects of Quantum Error Correction on Grover’s Algorithm using Shor’s 9-Qubit Code

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Introduction

Quantum computing suffers from high inaccuracy due to the hardware’s susceptibility to environmental noise. Quantum error correction is an expanding field that explores algorithmic ways to correct errors from decoherence due to noise. This research project attempts to examine one method of quantum error correction, known as Shor’s 9-qubit code. It is applied to Grover’s algorithm for unstructured search to determine if desired state is returned more frequently with error correction than without.

Algorithms

Grover’s algorithm is an unstructured search algorithm composed of two parts: the oracle to mark the desired state, and the diffuser to increase it’s probability of being returned. Figure 1 shows the implementation of Grover’s with 2 qubits used in this research to find the state 00.

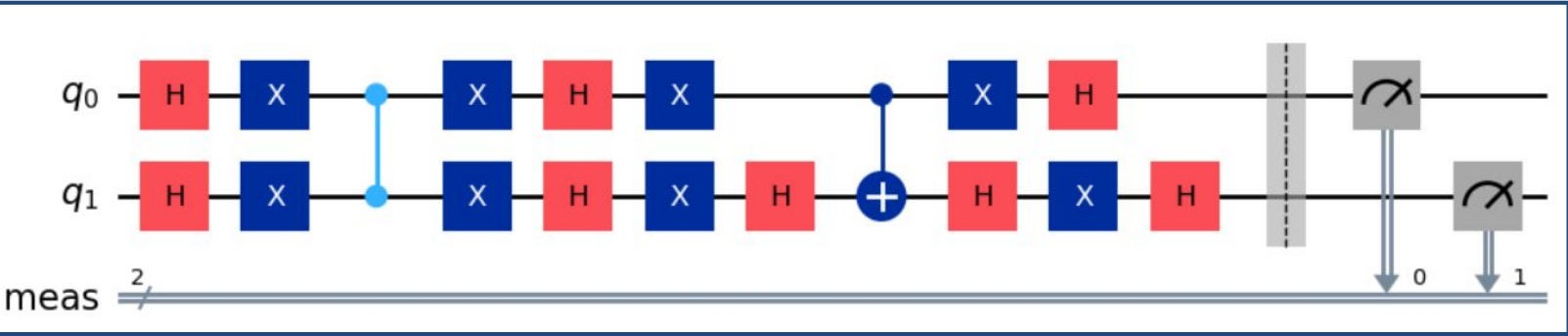


Figure 1

Shor’s 9-qubit code encodes 1 logical qubit across 9 physical qubits. An operation that can introduce error is performed across all 9 qubits, and the qubits are compared in groups of 3 to correct any errors that occurred, then are re-encoded back to the ancilla qubit. Figure 2 shows how 1 qubit can be encoded and decoded with Shor’s Code.

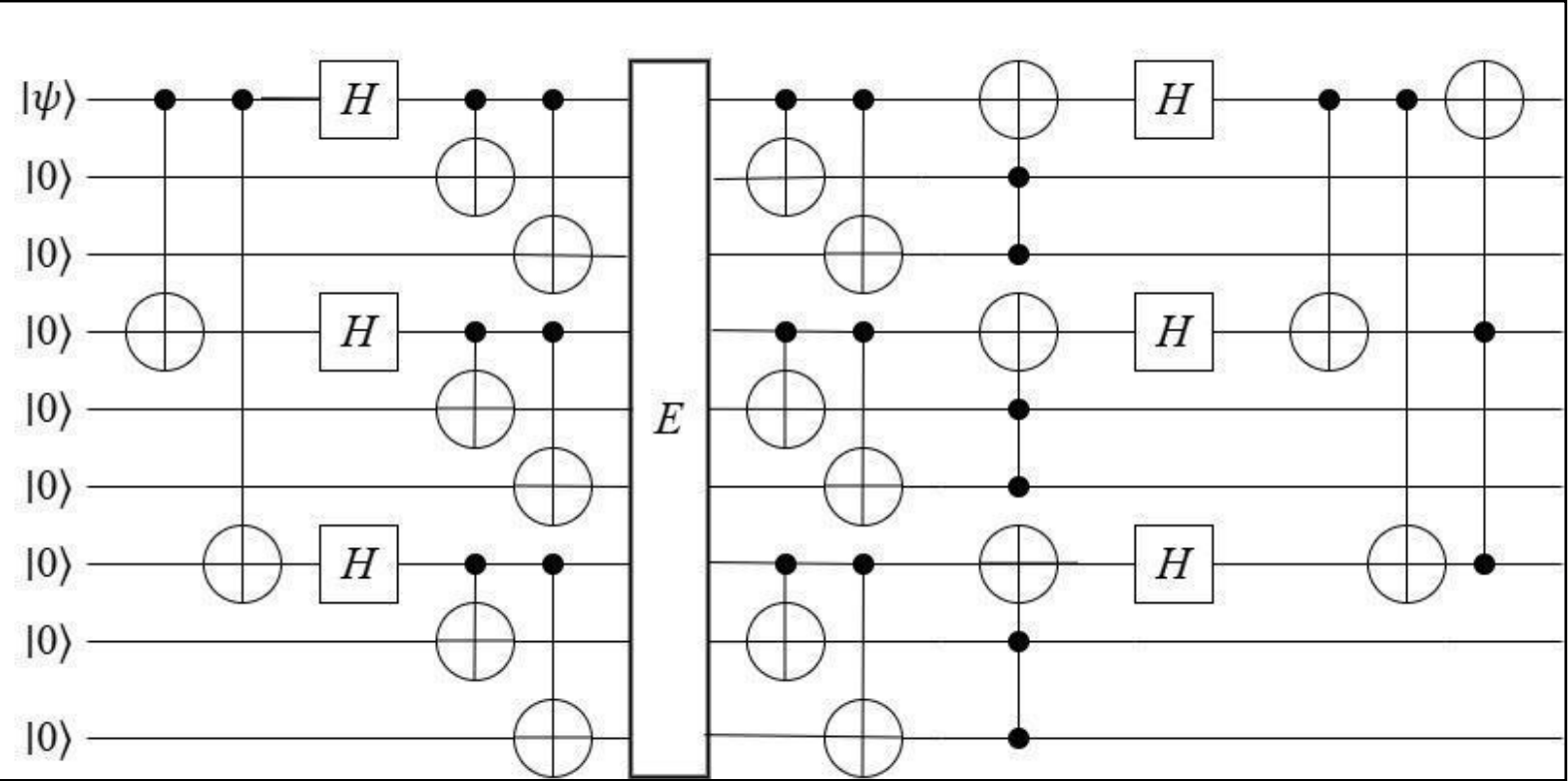


Figure 2

Methods

I developed Grover’s circuits with 2 and 3 qubits. They were run multiple times to search for each state.. I applied Shor’s code to each of these algorithms and ran them again. The frequency with which each state was returned was recorded in tables and graphed. Circuits were run on a both a noisy and noiseless simulator

Technologies



Results

2 Qubits - Noiesless Simulator		
Search State	Error Correction	Frequency
00	No	100.00%
	Yes	100.00%
01	No	100.00%
	Yes	100.00%
10	No	100.00%
	Yes	100.00%
11	No	100.00%
	Yes	100.00%

3 Qubits - Noiseless Simulator		
Search State	Error Correction	Frequency
000	No	76.86%
	Yes	33.59%
001	No	79.88%
	Yes	32.42%
010	No	79.10%
	Yes	32.32%
011	No	77.25%
	Yes	31.35%
100	No	78.90%
	Yes	29.69%
101	No	78.61%
	Yes	33.20%
110	No	78.22%
	Yes	31.05%
111	No	78.32%
	Yes	31.45%

2 Qubits - Noisy Simulator		
Search State	Error Correction	Frequency
00	No	88.78%
	Yes	98.44%
01	No	87.50%
	Yes	98.93%
10	No	88.18%
	Yes	98.54%
11	No	88.18%
	Yes	98.63%

3 Qubits - Noisy Simulator		
Search State	Error Correction	Frequency
000	No	74.90%
	Yes	27.34%
001	No	75.98%
	Yes	30.47%
010	No	75.59%
	Yes	27.34%
011	No	75.20%
	Yes	28.91%
100	No	78.91%
	Yes	25.00%
101	No	72.36%
	Yes	26.56%
110	No	73.73%
	Yes	26.56%
111	No	73.14%
	Yes	28.91%

Discussion

The error correction technique was effective for 2 qubits on the noisy simulator, increasing the chances of returning the marked state from about 88% to about 98%. However, it showed decreasing accuracy with 3 qubits on both simulators, decreasing from about 70-80% without Shor’s to around 30% with.

Conclusion

This research shows that Shor’s 9-qubit code may be an effective method to correct errors in Grover’s Algorithm. However, more research is needed to confirm these results and explore why it appears to become less effective as Grover’s scales up.

Full Work

All code used to produce this research can be found on my GitHub, along with a complete summary of the data, including all numbers and figures: <https://github.com/hannahtiffany3/CapstoneProject>
The preliminary research paper, published at the Computer Science Conference for CSU Undergraduates, is available here: <https://cscsu-conference.github.io/>